

Product Requirements Document

AI Room Reshuffle & Redesign App

Working name: ReSpace AI

An AI-powered room planning app that turns a room video into editable room understanding, reshuffle layouts, redesign concepts, fit-aware furniture suggestions, and step-by-step setup plans.

Document Type	PRD
Primary MVP	Reshuffle Existing Room
Secondary Expansion	Measurement-based Redesign + Product Recommendations
Core Differentiator	Video-based room planning with editable AI outputs and fit-aware suggestions

1. Product Overview

Product Vision

ReSpace AI helps users improve their living spaces without hiring an interior designer, manually measuring every wall, or struggling to imagine layout changes. Users record or upload a video of their apartment, room, or space. The system detects room structure, visible items, and estimated measurements, then generates practical reshuffle or redesign options that users can customize.

Core Product Promise

Users should be able to scan a room and receive actionable layout or redesign suggestions that are visually clear, editable, and grounded in the real dimensions and objects of the space.

Problem Statement

Many people want to improve their apartment or room but do not know how to arrange furniture effectively, cannot visualize layout changes, do not know whether new furniture will fit, and do not want to pay for an interior designer. Existing AI interior design tools often generate attractive images but ignore real item placement, measurements, budget, and fit constraints.

Product Opportunity

The strongest positioning is not generic AI interior design. The stronger product direction is: AI-powered room reshuffling and redesign from real room video, with editable measurements and fit-aware recommendations.

2. Goals and Non-Goals

Product Goals

- Help users reshuffle existing spaces using current furniture.
- Help users redesign spaces with new items in later versions.
- Build trust through user-correctable AI outputs.
- Generate practical outputs: layout explanation, moved items, fit reasoning, and setup steps.
- Reduce uncertainty before users move furniture or purchase products.

Non-Goals for MVP

- Full 3D room editor
- Real-time AR furniture placement
- Multi-room floor plan generation
- Designer marketplace
- Automated checkout
- Perfect measurement accuracy from video alone
- Professional CAD-level plans
- Complex commercial-space support

MVP Reality Check

The MVP should not attempt to solve reshuffling, redesign, exact measurement, product shopping, AR, 3D editing, and marketplace functionality all at once. The first version should focus on one complete workflow: upload or record room video, detect items, confirm items, choose goal, generate reshuffle layouts, and save final plan.

3. Target Users

Primary Segments

Segment	Needs
Apartment renters	Better furniture placement, more open space, small-space optimization, budget-friendly ideas
Students and young professionals	Functional study/work areas, low-budget options, easy reshuffle plans
New movers	Fast layout planning, setup checklist, product suggestions
DIY home improvers	Multiple layout ideas, decor suggestions, before/after visualization

Secondary Segments

- Interior design beginners
- Small landlords and Airbnb hosts
- Furniture shoppers who need fit confidence before buying

4. Product Modes

Mode A: Reshuffle Existing Room

Helps users rearrange existing furniture and items without buying new products. User intent: I already have stuff. Tell me how to arrange it better. Core output: 3-5 reshuffle layout images, top-down layout view, items moved, difficulty level, and step-by-step move instructions. MVP priority: High.

Mode B: Redesign Room

Helps users redesign a room using new products, style preferences, budget, and measurements. User intent: I want this room to look better, and I am open to buying new things. Core output: redesigned concepts, product recommendations, estimated budget, fit confidence, shopping list, and setup instructions. MVP priority: Medium to low for first release.

5. User Personas

Persona	Situation	Pain Points	Success Outcome
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Maya - Small Apartment Renter	Studio apartment; wants more space	Cluttered room, no designer budget, cannot visualize better layout	Receives three reshuffle options and follows a setup checklist
Daniel - New Mover	One-bedroom apartment; wants redesign	Unsure sofa size, fixed budget, wants modern look	Corrects measurements and gets product-fit redesign options
Priya - Work-From-Home User	Shared apartment; needs better bedroom office	Bad lighting, awkward desk placement, fixed bed/wardrobe	Locks fixed items and gets desk-near-window layouts

6. End-to-End User Flows

6.1 Reshuffle Flow

- User opens app and starts a new project.
- User selects Reshuffle My Room.
- App shows capture instructions.
- User uploads or records video.
- System processes video: upload, frame extraction, room/item detection, layout preparation.
- User reviews detected items and marks movable/fixed objects.
- User selects reshuffle goal and constraints.
- AI generates 3-5 layout options.
- User compares layouts and customizes through quick actions or natural language.
- User saves final reshuffle plan.
- App provides final image, top-down layout, and step-by-step move checklist.

6.2 Redesign Flow

- User selects Redesign My Room.
- User selects room type.
- App provides stricter capture instructions.
- User uploads or records video.
- AI detects room structure: walls, floor, doors, windows, furniture, empty zones.
- AI estimates measurements and clearly labels them as estimates.
- User reviews and corrects measurements.
- User optionally calibrates using one known dimension.
- User selects budget, style, colors, stores, must-have items, and existing items to keep.
- AI generates 3-5 redesign concepts.

- User reviews product recommendations with price, dimensions, store, fit status, and alternatives.
- User customizes products and design.
- App generates final redesign plan with shopping list and setup checklist.

7. Functional Requirements

Area	Requirement Summary
Authentication and User Accounts	Users can sign up, log in, save projects, reset password, and delete account. Guest mode may be used for early testing, but saving projects should require an account.
Project Creation	Users can create, name, save, reopen, rename, and delete projects. Each project stores mode, room type, media, detected items, measurements, preferences, generated outputs, and final selected design.
Video Upload and Capture	Users can record or upload MP4/MOV videos. MVP should limit videos to 30-60 seconds and enforce file size limits. Upload progress, validation errors, and retry states are required.
Capture Guidance	Before recording, the app must explain how to capture useful video: move slowly, good lighting, show floor/walls/corners, include major furniture, avoid fast movement, and optionally include known object for scale.
Object Detection	System detects core categories: sofa, chair, table, bed, desk, TV, TV stand, bookshelf, cabinet, wardrobe, lamp, rug, plant, door, and window.
Detected Item Review	Users can rename, delete, add, and mark items as movable or fixed. Layout generation must respect fixed items.
Room Measurement Estimation	For redesign mode, system estimates room width, length, wall height, door/window dimensions, furniture dimensions, and empty spaces. All AI measurements must be labeled estimated.
Calibration Step	User can enter one known measurement such as door width, bed size, desk width, A4 paper, credit card, or manual wall length. System recalculates estimates.
Preference Collection	Reshuffle preferences include goal, fixed items, desired feeling, function priority, and movement difficulty. Redesign preferences include budget, style, colors, stores, must-have items, keep/remove lists.
Layout Generation	System generates 3-5 options. Reshuffle outputs include layout image, top-down layout, goal alignment, moved items, unchanged items, difficulty, and explanation. Redesign outputs include render, product list, cost, fit confidence, and alternatives.
Design Customization	Users can regenerate, lock items, request changes, change style or budget, remove/replace/add products, and filter by store.

Product Recommendation	Product cards show image, name, store, price, dimensions, fit status, buy link, and alternatives. Products without dimensions should be labeled risky or unknown.
Final Plan Generation	Final plans include image, top-down layout, step-by-step checklist, moved/unchanged items, shopping list if redesign, total cost, and fit warnings.

8. Core Screens

Screen	Purpose	Main Elements
Welcome	Introduce app and start project	Logo, value proposition, Start New Project, Saved Projects
Mode Selection	Choose workflow	Reshuffle card, Redesign card
Capture Instructions	Improve video quality	Tips, good/bad examples, continue button
Upload / Record	Collect video input	Record, upload, file requirements, progress
Processing	Show analysis progress	Upload, frame extraction, detection, generation states
Detected Items Review	Correct AI detection	Item cards, rename/delete, movable/fixed toggle, add item
Measurement Review	Correct dimensions	Measurement table, edit fields, calibration, unit selector
Preferences	Collect goals/style/budget	Goal, style, budget, fixed item, store selectors
Results	Show generated options	3-5 cards, image preview, difficulty/cost, save, compare
Design Detail	Review one option deeply	Large image, top-down layout, explanation, customize
Products	Review furniture/decor	Product cards, price, dimensions, fit status, replace
Final Plan	Give actionable instructions	Final image, setup checklist, shopping list, save/export
Saved Projects	Manage projects	Project cards, room type, last modified, mode, actions

9. MVP Scope

MVP Version 1: Reshuffle Existing Room

Objective: Validate that users find value in uploading a room video and receiving useful layout suggestions using existing items.

Must Include

- New project creation
- Reshuffle mode
- Video upload
- Basic frame extraction
- Object detection
- Detected item review
- Preference selection
- At least 3 layout suggestions
- Save selected layout
- Final move checklist

Should Not Include

- Product recommendations
- Full redesign mode
- AR preview
- 3D editor
- Exact measurement claims
- Marketplace integrations

MVP Success Criteria

- Users complete the reshuffle flow without confusion.
- Users trust and use the detected item correction flow.
- Users save at least one generated layout.
- Users describe output as actionable, not just visually interesting.

10. Version Roadmap

Version	Focus
V1 - Reshuffle MVP	Upload/record room, detect items, confirm items, generate reshuffle layouts, save final plan
V2 - Measurement + Redesign	Estimate room dimensions, manual correction, calibration, style/budget preferences, redesign concepts

V3 - Product Recommendations	Furniture search, product dimensions, store filters, fit confidence, shopping list
V4 - Interactive Editing	Drag-and-drop editor, item locking, manual placement, top-down editing
V5 - AR and Advanced Visualization	AR preview, live overlay, 3D room model, real-time placement
V6 - Monetization and Marketplace	Subscriptions, credits, PDF export, designer review, affiliate shopping

11. AI and System Behavior Requirements

- AI must be editable: users can correct items, measurements, room type, fixed items, preferences, and budget.
- AI must explain layouts: why it works, what changed, what stayed, and tradeoffs.
- AI must respect constraints: fixed items do not move, budgets are respected, products that do not fit are not primary recommendations.
- AI must be honest about uncertainty: use estimated measurement, approximate fit, confidence score, and manual verification language. Avoid exact, guaranteed, or perfect claims.

12. Data Requirements

Data Type	Fields
User Data	User ID, name, email, auth provider, preferences, saved projects
Project Data	Project ID, user ID, project name, room type, mode, created/updated dates
Media Data	Video URL, extracted frame URLs, thumbnail, upload status, processing status
Detection Data	Item ID, name, confidence score, bounding box, movable/fixed status, correction status
Measurement Data	Room width/length, wall height, door/window dimensions, source, units
Design Data	Design ID, project ID, image URL, top-down URL, prompt, explanation, items moved, difficulty, rating
Product Data	Product ID, name, store, price, image URL, dimensions, product URL, fit status

13. Success Metrics

Metric Category	Examples
Activation	Project starts, video uploads, item review completions, layout generations

Engagement	Generated layouts viewed, customization rate, saved design rate, return-to-project rate
Quality	Layout rating, object correction rate, measurement correction rate, usefulness rating, product fit satisfaction
Business	Free-to-paid conversion, ARPU, affiliate click-through, product purchase conversion, premium generation purchases

14. Monetization Strategy

- Free tier: limited projects, limited generations, basic reshuffle suggestions.
- Paid tier: more projects, redesign mode, product recommendations, PDF export, high-quality renders, saved history.
- Credit model: extra generations, premium renders, product-matched redesigns, AR previews later.
- Affiliate revenue: Amazon, IKEA, Walmart, Wayfair, Target, Home Depot.
- Designer add-on: future human designer review of AI-generated plans.

15. Risks and Mitigations

Risk	Impact	Mitigation
Poor measurement accuracy	Users lose trust if products do not fit	Label estimates, user correction, calibration, fit confidence
Pretty but impractical layouts	Product becomes a toy	Top-down layout, move checklist, fixed-item constraints, tradeoff explanations
Missed object detection	Incomplete layouts	Item review, add missing items, manual marking
Unreliable product data	Wrong prices/availability/dimensions	Trusted APIs, last-updated status, dimension labels, avoid unknown-dimension primary picks
Scope gets too large	MVP becomes messy and slow	Build reshuffle first; delay AR, 3D, marketplace
Inconsistent image generation	Unrealistic room designs	Structured prompts, constraints, multiple options, regeneration and feedback

16. Edge Cases

- Bad video quality: warn for dark, shaky, incomplete, or unclear video.
- Empty room: suggest redesign mode or ask user to add planned items.
- Cluttered room: require manual confirmation and offer decluttering-focused layouts.
- Mirrors/reflections: warn they may affect detection accuracy.
- Pets/people: ignore temporary objects and people.

- Built-in furniture: allow cabinets, wall-mounted TVs, counters, and wardrobes to be fixed.
- Unknown product dimensions: label risky and avoid as primary recommendations.

17. Privacy and Safety Requirements

- Secure upload and private projects.
- Ability to delete videos, frames, projects, designs, and account data.
- Do not use user videos for training without explicit consent.
- Avoid analyzing or exposing sensitive personal details in videos such as documents, mail, screens, personal photos, and children faces.
- Consider deleting raw videos after processing unless user chooses to save them.

18. Accessibility Requirements

- Readable text and high contrast UI.
- Large tappable buttons.
- Screen reader labels.
- Simple language.
- Progress indicators and clear error states.
- Support feet/inches and meters/centimeters.

19. Technical Assumptions

- Video upload and storage.
- Frame extraction.
- Computer vision object detection.
- Room structure estimation.
- AI image generation.
- Layout generation.
- Product search integration.
- User project storage.
- Background processing and queue-based generation jobs.

20. Acceptance Criteria Summary

- User can create a reshuffle project.
- User can upload or record a room video.

- System detects basic room items.
- User can correct detected items.
- User can select reshuffle preferences.
- System generates at least three layout options.
- User can view and compare layouts.
- User can save one layout.
- App provides step-by-step reshuffle instructions.
- User understands AI outputs are suggestions, not guaranteed professional plans.

21. Open Questions

Product Questions

- Should the app start mobile-only or web plus mobile?
- Should users create an account before generating designs?
- Should MVP support only one room type first?
- Should pricing use per-generation credits or subscription?
- Should outputs be photorealistic, top-down, or both?

Technical Questions

- Which AI model handles object detection?
- Which model handles room layout understanding?
- Which model generates redesigned room images?
- How accurate can video-based measurement be without LiDAR?
- Should raw videos be stored or deleted after processing?
- Which product APIs or affiliate feeds should be used?

Business Questions

- Is the first target market renters, students, or homeowners?
- Will the app focus on budget or premium design?
- Should product recommendations monetize via affiliate links?
- Should human designer review become a premium feature?

22. Recommended Build Strategy

Phase	Goal
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Prototype	Accept images or short video frames and generate reshuffle concepts. Prove users understand the flow and find layouts useful.
MVP	Full reshuffle flow: upload video, detect items, edit items, select goals, generate layouts, save plan.
Redesign Beta	Measurement estimation, manual correction, style preferences, budget, redesign image generation.
Product Matching	Product recommendations, dimensions, store filtering, shopping list, fit confidence.
Premium Platform	Subscription, credits, PDF export, AR preview, human designer review.

23. Final Product Direction

The strongest version of this app is not a generic AI interior design generator. That market is crowded and many tools produce unrealistic outputs. The strongest direction is a practical AI room planning assistant that uses real room video, editable item detection, user-corrected measurements, and fit-aware layout/design suggestions to help users confidently improve their space.

The MVP should focus on practical reshuffling first. Redesign, shopping, AR, and marketplace features should come later after the core scan-to-layout experience proves useful.